SOLANO COMMUNITY COLLEGE · SUMMER 2026

BIO 004 Human Anatomy

Course syllabus for an 8-week summer intensive. A flipped, Team-Based Learning course in human anatomy with Dr. Sharilyn Rennie. Read it through, then keep it as your reference for the term.

COURSE	BIO 004, Human Anatomy
TERM	Summer 2026, 8-week intensive
DATES	June 15 to August 7, 2026
CLASS DAYS	Monday through Thursday
UNITS	5.0
FORMAT	In person, flipped TBL
PREREQUISITES	None
MEETING TIME AND ROOM	See the Canvas course homepage

What is in this syllabus

Instructor and contact

Your instructor for this course is **Dr. Sharilyn Rennie**, Professor of Biology at Solano Community College.

How to reach me and when to expect a reply.

EMAIL	sharilyn.rennie@solano.edu
PREFERRED CHANNEL	Canvas Inbox. It keeps all course messages in one place.
RESPONSE TIME	48 to 72 hours on weekdays. Plan ahead; do not expect a same-day reply the night before a deadline.
VIRTUAL OFFICE	Post a question to the "Virtual Office" board in Canvas. If you have a question, a classmate probably has it too.
OFFICE HOURS	By appointment. Message me through the Canvas Inbox to set a time.

> In an 8-week intensive, small problems become large problems fast. Reach out **before** you fall behind, not after. I would much rather hear from you in week 2 than in week 6.

Course description and outcomes

BIO 004 is a study of the structural organization of the human body, from the cellular to the organismal level. Throughout the course you will use several modes of investigation, including microscopic examination of prepared slides of tissues and organs, gross (macroscopic) anatomical study, and examination of prosected human material. (Formerly BIO 006.)

This course is pure anatomy. The focus is structure: what a structure is, where it is, what it looks like, and how it relates to the structures around it. Clinical context is threaded throughout so that the anatomy connects to real healthcare practice.

Student learning outcomes

On successful completion of this course, you should be able to:

1. Identify and distinguish tissues based on currently recognized criteria.
2. Describe the basic microscopic and gross anatomy of the major systems of the human body.
3. Identify, in the laboratory, the major microscopic and gross structures of the major systems of the human body.

Advisory: prior completion of BIO 16 (Introduction to Human Biology) and BIO 16L is recommended but not required.

How this course works

This is a flipped, Team-Based Learning (TBL) course. There are **no concept lectures during class time**. You learn the foundational anatomy on your own time through assigned videos, readings, and your weekly research workbook. Class time is for application: TBL discussion, case-based clinical reasoning, and hands-on lab work.

If you have taken courses where the professor lectures at the front while you take notes, this course will feel different. The difference is intentional. Active learning produces better mastery and retention than passive lecture, video is a better medium for content delivery because you can pause and rewind it, and class time is too valuable to spend on something you could do alone. In class you have a content expert and a team of peers in the same room. We use that time for the work you cannot do by yourself.

Your job

Show up prepared every class day. Complete your weekly research workbook before class. Watch the assigned lecture videos at your own pace. Engage actively during TBL. Use lab time for active recall, not socializing. Communicate with your team within 24 hours. Ask for help early.

My job

Curate and sequence the right content. Provide clear, structured lecture videos. Run TBL with rigor and consistency. Bring expertise to your application discussions. Design lab activities that build real skill. Assess your work fairly and quickly. Hold the standards healthcare programs expect.

The hard truth. If you cannot or will not do the foundation work outside of class, this course will not work for you. The flipped TBL model is built around prepared students. Do the work, and the system carries you. Skip it, and that gap will surface quickly.

For the full operating manual, see **How This Course Works** and the orientation page **Welcome to BIO 004**.

The weekly rhythm

Every week of this course rotates around the same four anchor activities: your **research workbook** (your weekly scaffold for foundation research), **Team-Based Learning** on Thursday, **lab work** every class day, and the **end-of-day lab quiz**. Once you know these four, the daily rhythm is just the order in which they happen. Class meets Monday through Thursday. Friday is a required rest day.

The shape of a typical week.

DAY	FOCUS	WHAT HAPPENS
Sunday	Prime the week	Open this week's research workbook. Watch assigned videos for Monday. Begin foundation research.
Monday	Class day	Lab practical exam on the prior week's content, then the new week opens. Case clue 1 released. Lab work. End-of-day lab quiz.
Tuesday	Class day	Case clue 2 released. Lab work continues. End-of-day lab quiz.
Wednesday	Class day	Case clue 3 released. Lab work and integration. End-of-day lab quiz. Workbook should be substantially complete tonight.
Thursday	TBL day	iRAT, then tRAT, then the case reveal and team application work. Lab work. End-of-day lab quiz.
Friday	Rest day	Required rest. Sleep is consolidation. No studying.
Saturday	Exam prep	Self-test for Monday's lab exam. Atlas drilling under time pressure. Track weak spots.

The cycle is the curriculum. Follow this rhythm consistently for 8 weeks and you will earn a strong grade. Compress the work into the day or two before each assessment and the system will surface that gap quickly.

Course materials and technology

Textbook (recommended, not required)

No textbook is required. Pick the option that fits your budget and learning style. Any of these covers the anatomy we will study.

- **OpenStax Anatomy and Physiology, 2nd Edition.** Free, peer-reviewed, available online and as a free PDF at openstax.org. The most accessible option and sufficient for the course.
- **Marieb, Brady, and Mallatt, Human Anatomy, 9th Edition.** ISBN-13: 9780135240601. Used copies are widely available. This is the dedicated anatomy text.
- **Marieb, Human Anatomy and Physiology**, any recent two-semester edition. If you already own this, it works.

Mastering A&P access is optional and is not required for any graded component. Do not buy it just because it is bundled with a textbook.

Lab supplies (needed by week 3)

- Non-latex nitrile gloves. Share a box with your group to reduce cost.
- A lab coat (required). It is stored in the lab through the end of the term and cannot be shared with chemistry. You need your own.

> **Order early.** The 8-week pace means you need lab supplies on hand by week 3, not week 6. Plan to order or pick them up during week 1 or 2.

Required technology

You must have access to a tablet, laptop, or desktop able to run multiple browser tabs and stream video; a reliable internet connection; a printer and scanner or a scanning app; a camera-enabled device; and a way to convert files to PDF. All scanned or photographed work is submitted as PDFs through Canvas. You are responsible for ensuring your devices meet these requirements.

Full detail, including optional supplementary materials, is on the Course Materials page.

Grading and assessment

Your course grade is built from four components. Point values may vary across the term, but the final grade is weighted to these percentages.

Grade weighting by component.

COMPONENT	DETAIL	WEIGHT
Team-Based Learning (TBL)	8 weekly TBLs, each with an iRAT and a tRAT. Every TBL counts. No scores are dropped.	**40%**
Lab Practical Exams	8 lab practical exams, one covering each week's content.	**35%**
Cumulative Lecture Final	One comprehensive lecture final at the end of the term, covering all modules.	**15%**
End-of-Day Lab Quizzes	Community-accountability quizzes given at the end of nearly every lab session.	**10%**

> **How the final protects your TBL grade.** There are no dropped TBL scores this term. Instead, your Cumulative Lecture Final does double duty. If your final score is higher than your single lowest TBL score, that lowest TBL is raised to match your final. Preparing well for the final can repair one weak TBL week. It can only help you; it never lowers a score.

Team-Based Learning (TBL)

There is one TBL each week, completed Thursday morning with your assigned team. Each TBL has two graded parts:

- **iRAT** (individual readiness assessment): a 10 to 15 question quiz you take on your own. Weighted 3x, worth 30 points.
- **tRAT** (team readiness assessment): the same quiz, taken again with your team. Weighted 1x, worth 10 points.

Each TBL is worth 40 points total, so the iRAT alone is 75% of your TBL grade. Most teams score 90% or higher on the tRAT, which means your individual preparation is what determines your TBL outcome. A strong team will not save an unprepared individual. After the tRAT, you and your team work through clinical application questions and the case reveal. Application questions are not graded, but you are expected to understand the reasoning if called on. There are no make-ups for a missed TBL.

Lab practical exams

There are eight lab practical exams. Exams 1 through 7 run on Monday mornings of weeks 2 through 8, each covering the prior week's content with roughly 20% spiral review of earlier material. Exam 8 runs on the Thursday afternoon of week 8 and covers week 8 content. Every lab exam counts; none is dropped.

Format

- 50 minutes, 25 stations, 2 questions per station, for 50 questions total.
- 90 seconds per station. You rotate on a timer.
- You identify anatomy from microscope slides, pictures, charts, models, and prosected material.
- Spelling counts. Bring your favorite pen. Pencils are not permitted.

Conduct

- Doors lock when the exam begins. If you are late, you cannot enter.
- Do not look around the room or at other stations. Fold your paper to cover your answers.
- Failure to follow exam instructions results in collection of your exam, graded as is.
- No make-ups.

Cumulative Lecture Final

The Cumulative Lecture Final is one comprehensive exam given at the end of the term, in week 8. It includes equal representation of material from every module and is weighted at 15% of your course grade. As described above, it also serves as the safety net for your lowest TBL score. The exact date and time are posted on the Canvas course homepage.

End-of-day lab quizzes

At the end of nearly every lab session, I ask one person in one randomly selected group a single question covering that day's lab content. The whole class earns the same score based on that one answer. Each quiz is worth 3 points. Students who are present earn the class score; students who are absent earn a zero, with no make-ups. No lab quiz scores are dropped. The format is intentional: your lab quiz grade depends on the readiness of everyone in the room, so use lab time well and show up for each other.

Depth of Knowledge (DOK)

Assessment questions are written at different Depth of Knowledge levels and are designed to prepare you for healthcare applications, not just memorization. Most questions are DOK 2 and DOK 3.

- **DOK 1**, basic recall (about 10%): naming structures, defining terms.

- **DOK 2**, skills and concepts (about 40 to 50%): applying knowledge, identifying relationships, interpreting images.
- **DOK 3**, strategic thinking (about 40 to 50%): solving clinical problems, justifying answers, predicting impact.
- **DOK 4**, extended thinking (about 0 to 10%): bonus or optional integrative work.

You can estimate your standing at any point with the **Grade Calculator**, and the full assessment detail lives on the **Grading and Assessments** page.

Grading scale

A	B	C	D	F
89.5 to 100%	79.5 to 89.4%	69.5 to 79.4%	59.5 to 69.4%	0 to 59.4%

This is a mastery-based course. Attending class and completing assignments does not by itself guarantee success. Strong performance requires active engagement and demonstrated understanding. Plan for roughly 20 or more hours of work per week outside of class to earn a passing grade in this intensive format.

Scholar Points (up to 3.0% bonus)

Scholar Points reward proactive learning and academic competence. They are not extra credit for missed work; they recognize consistent effort and performance across the term.

The two requirements for the 3.0% bonus.

REQUIREMENT	DETAILS	BONUS
Tutoring engagement	18 verified hours of approved on-campus tutoring across the term. A maximum of 3 hours per week is counted, which spreads tutoring across at least 6 weeks.	**2.0%**
Academic performance *(only with tutoring)*	Maintain an average of 80% or higher across lab exams, lab quizzes, and TBL. This 1% is awarded only if you also complete the 18 tutoring hours.	**1.0%**

You may also earn up to 25 supplemental Scholar Points from end-of-day lab challenge questions and from hosting or attending Community Rounds. The maximum bonus is 3.0%, applied at the end of the term.

> **No other extra credit will be given in this course.** Scholar Points are the only path to any extra credit. Please do not ask. No exceptions.

Course schedule

The course runs eight weeks, June 15 to August 7, 2026, with class Monday through Thursday. Each week pairs a body system with a clinical case that your team works up live and presents at Thursday's TBL. We will try to follow this schedule, but dates and point values can change; if you miss a class, check with your team for any announced changes.

Week-by-week overview. Week numbers link to the full weekly page.

WEEK	MODULE AND ANATOMY FOCUS	LAB EXAM
Week 1 **Jun 15–18**	**Module 1.** Foundations and Integumentary System. Anatomical terminology, body cavities, cells and tissues, blood as a fluid connective tissue, skin and appendages.	None this week
Week 2 **Jun 22–25**	**Module 2, Part 1.** Skeletal: bone microanatomy and the axial skeleton. Bone classification, ossification, skull, vertebral column, thoracic cage.	Lab Exam 1, Mon Jun 22 (covers Week 1)
Week 3 **Jun 29–Jul 2**	**Module 2, Part 2.** Appendicular skeleton, joints, and muscle as a system. Girdles and limbs, joint classification, synovial joints and movements.	Lab Exam 2, Mon Jun 29 (covers Week 2)
Week 4 **Jul 6–9**	**Module 3, Part 1.** Cardiovascular: heart, great vessels, thoracic and upper extremity region. Heart anatomy and conduction, vessel histology, thoracic and arm muscles.	Lab Exam 3, Mon Jul 6 (covers Week 3)

WEEK	MODULE AND ANATOMY FOCUS	LAB EXAM
Week 5 **Jul 13–16**	**Module 3, Part 2.** Respiratory, lymphatic, immune, and blood. Airways and lungs, pulmonary circulation, lymphatic organs, blood composition.	Lab Exam 4, Mon Jul 13 (covers Week 4)
Week 6 **Jul 20–23**	**Module 4.** Digestive system with abdominal, pelvic, and lower extremity regional anatomy. GI tract and accessory organs, abdominal vessels, lower limb muscles.	Lab Exam 5, Mon Jul 20 (covers Week 5)
Week 7 **Jul 27–30**	**Module 5.** Urinary, reproductive, and endocrine systems. Kidney and nephron, renal vasculature, reproductive systems, endocrine glands.	Lab Exam 6, Mon Jul 27 (covers Week 6)
Week 8 **Aug 3–6**	**Module 6.** Nervous system with head, neck, face, and eye regional anatomy. Brain and spinal cord, meninges and ventricles, cranial nerves and ANS. Cumulative Lecture Final this week.	Lab Exam 7, Mon Aug 3 (covers Week 7); Lab Exam 8, Thu Aug 6 afternoon (covers Week 8)

> **Holiday notes.** No classes the Juneteenth weekend of June 19 to 21. Independence Day is observed Friday July 3 (campus closed; Friday is already a rest day). The last day of the term is Thursday August 6.

The living schedule, which highlights the current week automatically, is the Course Schedule page.

Attendance and engagement

In an 8-week intensive, attendance is not just encouraged, it is essential. The cumulative nature of anatomy makes it very difficult to recover after missing class, because in this format one missed day is the equivalent of a full week in a regular semester.

- You must be present for **every class meeting during the first week**. If you are absent during the first week, your seat may be given to the next student on the waitlist and you may be dropped.
- You may miss a maximum of **3 class sessions** across the term, excused or unexcused. On the third absence, the instructor reserves the right to drop you from the course.

- You must be present for at least 90% of a session to be counted. Arriving 20 minutes late or leaving 20 minutes early counts as an absence.
- Missed in-class work cannot be made up. Missed TBLs, exams, lab quizzes, and activities earn a zero.
- Absences are treated uniformly. There is no excused versus unexcused distinction; you never need to explain an absence, and the academic consequence is the same either way.
- Doors lock once a lab exam starts. If you are late, you cannot enter.
- Log in to Canvas at least three times per week and set up notifications, at minimum for announcements.

Engagement is part of your grade. You will be placed on a TBL team for the term and are expected to come prepared, contribute, and respond to team communication within 24 hours. Your grade depends on both your individual preparation and your contribution to your team.

It is your responsibility to formally drop the course if you decide not to continue. Do not assume the instructor will do it for you. Failing to drop can result in an avoidable F.

Academic integrity and AI

Academic dishonesty, cheating, and plagiarism will not be tolerated. A violation results in at least a zero on the work in question and may result in an F for the course, and all violations are reported for disciplinary action. Complete your own work and cite your sources in MLA or APA format. If you need help with citation, ask.

Use of AI

You will be taught effective, ethical ways to use AI as a learning tool in this course, and I encourage you to use it for tutoring yourself and building study materials. For graded work, the rules are strict:

- You may use generative AI **only** on assignments specifically designated as AI assignments (the title will include "AI" or the instructions will say so).
- You may **not** use AI on exams or on any graded portion of a TBL activity.
- AI detection tools may be used to screen submitted work. An AI or plagiarism rating above 25% on a graded assignment earns an automatic zero.
- If you are unsure whether a tool counts as generative AI or is permitted on a given assignment, ask before you submit.

See the AI Honor Contract in Canvas for full requirements.

Make-up and late work

All quizzes, exams, lab work, and assignments are due by their scheduled dates. **No late work is accepted and there are no make-ups.** All in-class assessments must be completed in person; if you are absent, you earn a zero. Many assignments in this course are recommended but not graded; treat them seriously anyway, because they are designed to build the mastery the graded assessments measure. The key to this policy is simple: do not miss class.

Digital device policy

The use of digital devices in the anatomy laboratory is strictly regulated. We work with cadaveric human donors under a legal agreement with UCSF's body donation program. The people who donated their bodies to your education trust us to protect their privacy and dignity, and they cannot consent to being photographed or recorded.

- No phones, smart watches, AI glasses, tablets, laptops, or recording devices are permitted in the lab at any time.
- Devices must be silenced and kept fully out of sight, not in a pocket, lap, or on the table.
- No digital note-taking, photography, video, or audio recording in the lab.

Every student must read, sign, and comply with the full Digital Device Policy as a condition of enrollment. Violations may result in immediate removal from the course and further disciplinary action. Students under 18 must sign the policy, and a parent or legal guardian must also sign, before participating in lab. Read and complete the agreement here: **Digital Device Policy and Student Agreement**.

Your TBL team

There are three layers of peer learning in this course, and you should use all three. Your **TBL team** is your assigned home base: 5 to 7 students, balanced by a brief week 1 survey, kept the same for all 8 weeks. Your team is graded together on the tRAT and works application questions together. **Study groups** are voluntary groups you form to drill structures and run cases outside class. **Community Rounds** are peer-led, class-wide review sessions that you organize for each other and that earn Scholar Points.

By the end of class on the first day, your team must exchange contact information and agree on one communication platform. The standing expectation is a 24-hour response time. If your team is genuinely not working, contact me early, through Canvas or email, before the situation festers. Full guidance is on the **Your TBL Team and Study Groups** page.

Accessibility and accommodation

Your success in this class is important to me. This course is designed to be welcoming to, accessible to, and usable by everyone. If any aspect of the course prevents you from learning or excludes you, please let me know as soon as possible and we will develop strategies together.

Solano Community College's Accessibility Services Center (ASC) provides accommodations such as extra time on tests and specialized technology. ASC is on the Fairfield campus in Building 400, Room 407. Phone (707) 864-7136, email ASC@solano.edu, accommodations portal at clockwork.solano.edu.

> **Set this up in week 1.** In an 8-week intensive, a two-week delay in arranging accommodations is a quarter of the course. Contact ASC as early as possible, then inform me of your approved accommodations during the first week through the Canvas Inbox.

More detail is on the **Accessibility and Accommodation** page.

Where to get help

- **Your instructor.** Canvas Inbox or sharilyn.rennie@solano.edu, with a 48 to 72 hour response on weekdays.
- **Your TBL team.** Your first line of support. Use your team chat.
- **On-campus tutoring.** Embedded tutors for this course, with names provided in week 1. Also the Academic Success and Tutoring Center.
- **Office hours.** By appointment; message me to schedule.
- **Community Rounds.** Peer-led review sessions, self-organized and posted in the Canvas thread.
- **Canvas Support.** For technical issues with login, assignments, or video playback.

If you are ever disoriented in the term, return to **How This Course Works**. The rhythm is the cure for the disorientation of an intensive course.

Course pages and tools

Every page built for this course is linked below. Each opens in this same window. Bookmark the ones you will use most: this syllabus, the schedule, and How This Course Works.

Orientation and reference

Welcome to BIO 004 →

The headline orientation: what a flipped TBL course is and what to expect.

How This Course Works →

The operating manual: weekly rhythm, the research workbook, the four anchors.

Your TBL Team →

Team formation, the three layers of community, communication norms.

Course Materials →

Textbook options, lab supplies, and technology requirements.

Digital Atlas Guide →

How to use the digital atlas for active recall, not passive scrolling.

All Tools and Links →

The full course hub, including study tools and external resources.

Grading, schedule, and policies

Grading and Assessments →

Full assessment detail and the grading structure.

Grade Calculator →

Estimate your current standing from your component averages.

Course Schedule →

The week-by-week schedule, highlighting the current week.

Accessibility →

ASC contact, registration steps, working with accommodations.

Digital Device Policy →

The no-devices-in-lab rule, the reasons, and the signed agreement.

Spiral Rounds Procedure →

The five-round lab study procedure to run with your group.

Weekly pages

Each week has an overview page, a lab sprint page, and (for selected weeks) a clinical reasoning case simulator.

Week overviews

Week 1 | Week 2 | Week 3 | Week 4 | Week 5 | Week 6 | Week 7 | Week 8

Lab sprints

Lab 1 | Lab 2 | Lab 3 | Lab 4 | Lab 5 | Lab 6 | Lab 7 | Lab 8

Clinical reasoning case simulators

Case 1 | Case 2 | Case 8

Important dates

Key dates for Summer 2026. Confirm add, drop, and withdrawal deadlines on the Solano CC academic calendar.

DATE	EVENT
Mon Jun 15	First day of the term
Jun 19–21	Juneteenth weekend, no classes
Mon Jun 22	Lab Exam 1
Mon Jun 29	Lab Exam 2
Fri Jul 3	Independence Day observed, campus closed
Mon Jul 6	Lab Exam 3
Mon Jul 13	Lab Exam 4
Mon Jul 20	Lab Exam 5
Mon Jul 27	Lab Exam 6
Mon Aug 3	Lab Exam 7

DATE	EVENT
Week of Aug 3	Cumulative Lecture Final (exact date and time in Canvas)
Thu Aug 6	Lab Exam 8 and last day of the term

Add, drop, refund, and withdrawal ("W") deadlines are set by the college. Check the Solano CC academic calendar and your Self-Service portal at welcome.solano.edu, and remember that formally dropping the course, if you choose to leave it, is your responsibility.

A final word

This is going to be a demanding summer, and I will not pretend otherwise. But the 8-week intensive has a real advantage: you are in class four days a week, working the same material with the same team every day, and there is no time to drift. Many students perform better in this format than in the 16-week version of the same course. Show up, do the foundation work, lean into your team, and treat each other well. By the end of the term your team will feel like family, and that is by design. Your grade is feedback, not a verdict. Keep showing up, and keep going. I am rooting for you.